

HOW INTERNET WORKS

CSE 1110 – Introduction to Computer Systems

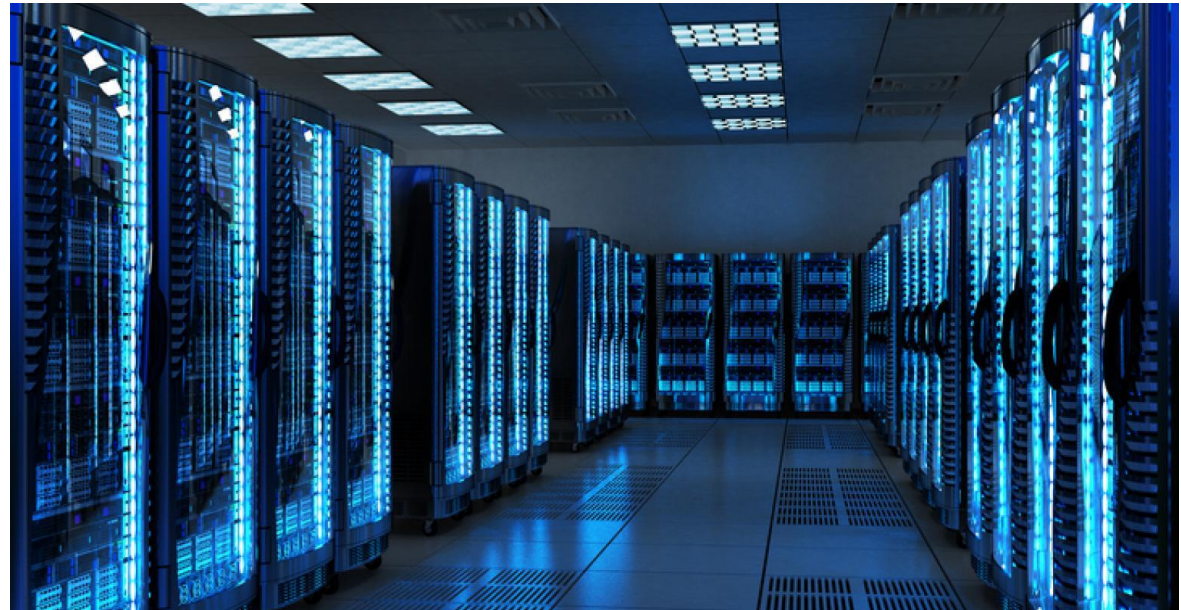
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Courtesy: Minhajul Bashir

THE BIGGEST NETWORK



CONNECTS ALL THESE



HOW INTERNET WORKS

All computers, smartphones, servers etc. are connected to this network

This network contains different types of links

- Wired link
- Wireless link
- Optical fiber link
- Satellite link

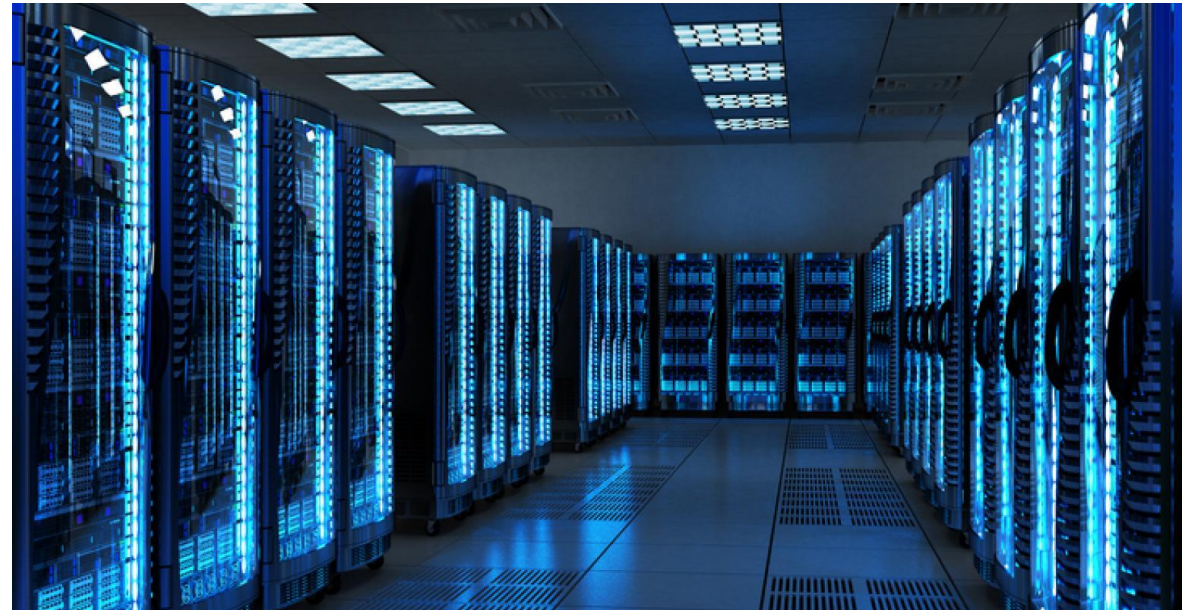
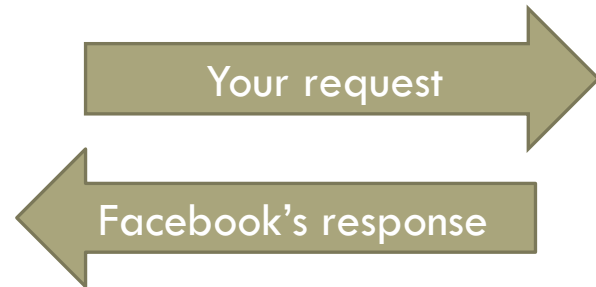
HOW INTERNET WORKS

When you enter “www.facebook.com” in your browser, you are requesting Facebook to show you your homepage

This request travels from your PC/phone, over the internet, to Facebook’s server

Facebook processes this request, gathers data, and sends you the reply over the internet

HOW INTERNET WORKS



NETWORKING DEVICES

Internet is a complex network, containing many paths

Networking devices in the internet determine which way to go to find Facebook's server

- Like railway junctions!!!

Example networking devices: switches, routers



IP ADDRESSES

Every single entity in the internet has a unique address called the IP address

Sample IP address: 192.178.54.15

The Facebook homepage request is sent from your PC/phone to Facebook's IP address

Facebook's response is sent to your PC/phone's IP address

QUICK QUIZ: WHAT IS THE IP ADDRESS OF FACEBOOK.COM?

I don't know either :P

DOMAIN NAME SERVER (DNS)

When you enter “www.facebook.com” in your browser, the first request goes to a Domain Name Server (DNS)

In response, DNS sends your browser the IP address of Facebook’s server

Now the browser knows where to send your request

How do you know the IP address of a DNS server?

- Provided by your ISP

DOMAIN NAME SERVER (DNS)

Website	IP address*
...	...
facebook.com	234.123.46.159
google.com	159.147.201.4
uiu.ac.bd	124.128.185.24
...	...

DNS works like a telephone directory

*These IP addresses are imaginary, and possibly do not represent any real website

